

LEMIEUX, QC, Canada

WMO: 716160

Lat: 46.300N

Lon: 72.060W

Elev: 97

StdP: 100.16

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.5	(c) -25.9	(d) -33.4	(e) 0.2	(f) -28.7	(g) -29.9	(h) 0.2	(i) -25.2	(j) 9.7	(k) -4.7	(l) 8.7	(m) -4.9	(n) 1.0	(o) 180	(p) 0.581

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	29.3	21.5	27.8	20.4	26.3	19.6	23.2	27.3	22.1	25.7	21.1	24.7	4.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.8	16.7	25.4	20.9	15.7	24.3	19.9	14.7	23.2	69.5	27.4	65.3	25.7	61.8	24.8	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.4	7.4	6.5	-35.3	31.9	4.3	1.3	-38.4	32.8	-41.0	33.6	-43.4	34.3	-46.5	35.2
			-34.8	24.8	4.1	1.2	-37.7	25.7	-40.2	26.4	-42.5	27.1	-45.5	28.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.9	-11.9	-10.2	-3.9	4.2	11.8	16.5	19.1	17.7	13.7	7.3	0.4	-6.8
	DBStd	11.98	8.09	7.13	6.54	4.50	4.21	3.88	3.08	3.51	4.27	4.40	5.38	6.80
	HDD10.0	2819	678	566	431	182	27	2	0	1	12	107	291	522
	HDD18.3	5027	937	799	689	425	207	80	27	53	149	342	539	780
	CDD10.0	962	0	0	0	7	83	198	283	241	123	24	2	0
	CDD18.3	128	0	0	0	0	4	26	52	35	11	0	0	0
	CDH23.3	1253	0	0	0	6	97	289	471	297	91	2	0	0
	CDH26.7	261	0	0	0	1	22	71	104	48	14	0	0	0
(14) Wind	WSAvg	2.9	3.1	3.2	3.3	3.3	3.1	2.7	2.5	2.4	2.5	2.9	3.0	2.9
(15) Precipitation	PrecAvg	1102	81	65	74	80	95	102	117	108	95	105	87	96
	PrecMax	1288	157	120	140	164	181	192	185	162	181	186	136	218
	PrecMin	902	38	30	25	22	45	42	47	61	55	36	25	36
	PrecStd	109	28	22	27	35	34	37	38	24	28	38	24	38
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.9	7.4	13.3	22.8	29.4	30.9	31.6	29.9	28.8	23.1	17.5	9.5
		MCWB	8.0	4.6	7.8	14.3	18.5	22.1	23.3	22.6	22.0	17.0	14.0	8.2
	2%	DB	5.1	4.5	9.4	18.4	26.0	28.8	29.3	28.0	25.9	19.9	13.4	5.7
		MCWB	4.0	2.3	5.2	10.8	16.1	20.6	22.1	21.0	20.3	14.7	10.8	4.9
	5%	DB	2.3	2.4	6.9	15.0	23.4	26.8	27.6	26.5	23.7	17.1	10.6	3.5
		MCWB	1.3	1.1	3.7	9.0	15.3	19.2	20.9	20.2	18.6	13.5	8.4	2.5
	10%	DB	-0.2	0.8	4.8	12.0	20.7	24.6	26.0	24.9	21.5	14.6	7.9	1.7
		MCWB	-0.9	-0.3	2.3	6.9	13.9	18.2	20.0	19.5	17.7	11.8	6.3	0.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.0	5.8	9.9	15.8	21.1	23.8	24.9	23.9	23.1	18.3	14.5	8.5
		MCDB	8.4	6.7	12.1	22.1	26.3	29.0	29.6	28.3	27.5	21.6	16.3	9.6
	2%	WB	3.9	2.6	6.2	12.2	18.7	22.1	23.5	22.5	21.4	15.9	11.5	5.2
		MCDB	4.6	3.9	8.3	16.6	23.3	26.4	27.4	26.0	24.5	19.0	13.2	5.9
	5%	WB	1.1	1.2	4.1	9.6	16.7	20.6	22.3	21.5	19.9	13.9	8.7	2.7
		MCDB	1.8	2.3	6.4	14.3	21.0	24.8	25.9	24.9	22.9	16.5	10.1	3.4
	10%	WB	-1.2	-0.2	2.6	7.6	14.9	19.4	21.2	20.6	18.4	12.2	6.4	1.1
		MCDB	-0.5	0.8	4.5	11.3	19.2	23.2	24.7	23.9	21.3	14.1	8.0	1.6
(35) Mean Daily Temperature Range	MDBR		11.2	12.4	11.1	10.7	12.9	12.5	11.8	12.1	12.2	9.9	8.3	8.8
	5% DB	MCDBR	11.3	11.5	12.6	17.0	17.7	15.4	13.5	13.6	14.0	14.0	12.2	7.9
		MCWBR	10.7	10.2	8.9	10.2	9.1	8.0	7.2	7.5	8.5	9.3	9.6	7.8
	5% WB	MCDBR	10.5	10.5	11.2	15.0	13.8	12.8	11.4	11.3	12.3	12.3	10.9	8.3
		MCWBR	10.3	9.5	8.4	9.9	8.4	7.4	6.5	6.9	8.1	8.9	9.3	8.1
(40) Clear-Sky Solar Irradiance	taub		0.262	0.282	0.315	0.343	0.367	0.406	0.412	0.399	0.368	0.332	0.304	0.282
	taud		2.341	2.260	2.296	2.395	2.393	2.275	2.273	2.314	2.394	2.497	2.490	2.343
	Ebn at Noon		845	893	911	914	899	861	850	847	845	831	792	769
	Edh at Noon		73	98	111	111	115	130	129	119	101	79	65	67
(44) All-Sky Solar Radiation	RadAvg		1.42	2.31	3.58	4.46	5.08	5.58	5.51	4.82	3.80	2.21	1.31	0.96
	RadStd		0.20	0.27	0.31	0.43	0.53	0.39	0.50	0.26	0.32	0.18	0.17	0.09

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page